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Local Metadata Catalogues in Medical Research: Bridging Interdisciplinary Standards Gaps to Reach Compatibility with NFDI and EHDS Requirements

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Abstract

Background. The development of the European Health Data Space (EHDS) [1] presents many opportunities and challenges for findable, accessible, interoperable and reusable (FAIR) data across the healthcare domain. At the beginning of 2025, TEHDAS2, the joint action Towards the European Health Data Space [2], shared the first public draft of the guidelines for national metadata health catalogues. However, addressing these guidelines across diverse research communities (e.g. epidemiology, radiology, pathology) remains complex, as each community has unique needs and requirements. Existing national initiatives (Medical Informatics Initiative, NFDI4Health) successfully work towards developing necessary solutions and standards [3,4,5], however, interdisciplinary harmonized schemas and software solutions for supporting local research teams are a work in progress.

Methods. Two teams, from the University Hospitals of Bonn and Cologne, involved in supporting interdisciplinary medical research and research infrastructure development, organized a half-day focused workshop on-site to discuss the challenges of respective local metadata catalogue development. Both teams were represented by professionals of different backgrounds: data analysts, software engineers, AI experts, project managers and data stewards. The workshop was structured as follows: a) free exchange of experiences with supporting interdisciplinary medical teams in RDM processes, with a specific focus on metadata aggregation and exchange; b) identification of an interdisciplinary use case relevant to all participants and high-level requirements for metadata catalogue; c) listing the software solutions for research metadata catalogues, tested by workshop participants and sharing relevant experiences; d) combining the first iteration of the requirements with the lessons learned from the software testing, to iterate more detailed requirements for a suitable metadata catalogue solution, as well as list the major encountered problems and desiderata for the context of the model use case.

Findings. The teams agreed to focus on the use case of the pre-clinical interdisciplinary research group, with individual PIs representing research domains with internally well-established data sharing standards and metadata schemas. For this use case, the establishment of a metadata catalogue has a twofold purpose: sharing detailed and possibly sensitive metadata within the team, and sharing project-level metadata openly via national

and international metadata catalogues. We identified the following points which call for improvement of the existing software solutions: a) an efficient data management and harmonization strategy to support integrative findability and accessibility, b) automatic preparation of the metadata on the study level for sharing with existing (national and international) open metadata catalogues. A collaborative review of the existing software solutions showed that while many important requirements are met, choosing a single tool that handles the local and national level requirements innately and provides an optimized interdisciplinary user experience is difficult.

Conclusions. Our findings highlight a gap between the development of high-level metadata guidelines and schemas (e.g., EHDS, TEHDAS) and specialized domain-specific schemas, which hinders seamless data integration and interoperability. This gap propagates to the related bottlenecks in the open software solutions suitable for “out-of-the-box” metadata handling. To bridge this gap, a collaborative approach is needed, involving both specialized communities and high-level policymakers, working on a clear roadmap for metadata mapping that ensures not only interoperability across diverse data sources but also navigates the development of the existing software solutions towards improved practical usability.

Keywords: metadata catalogues, EHDS, NFDI, medical research, FAIR, interdisciplinary

Author contributions

Conceptualisation: all co-authors, Investigation: all co-authors, Methodology: all co-authors, Writing – original draft: EK; Writing – review & editing: all co-authors

Competing interests

The authors declare that they have no competing interests.

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